AI-Generated Image Detection

A Comparative Study of CNN Architectures

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Problem & Motivation

The Deepfake Challenge

- Synthetic media misrepresented as authentic
- Growing threat to digital integrity
- Generated using GANs
- Enables fraud and misinformation

Why CNNs?

- Excel at visual pattern recognition
- Detect subtle AI artifacts
- Transfer learning leverages pretrained features

Our Approach

Comparative Study of 3 Architectures:

1. Custom CNN - Baseline
2. ResNet50 - Transfer learning
3. EfficientNet-B2 - Advanced

Dataset

- 152,710 images (Hugging Face)
- Real vs AI-Generated
- 80/10/10 train/val/test split

Dataset & Methodology

Dataset Characteristics

- **Source:** Hugging Face
- **Total:** 152,710 images
- **Classes:** Real (0) vs AI-Generated (1)
- **Split:** 122,168 train / 15,271 val / 15,271 test
- **Balance:** Equal class distribution
- **Diversity:** Portraits, animals, objects, scenes

Preprocessing

- Resize to 224×224 pixels (RGB)
- ImageNet normalization
- Data augmentation:
  - Random horizontal flips
  - Random rotation ($\pm 10°$)
  - Color jittering

Training Configuration

Parameter	Value
Hardware	A100 GPU
Precision	AMP (Mixed)
Batch Size	32
Workers	2
Test Set	15,271 images

Key Decisions

- Balanced dataset reduces bias
- ImageNet normalization for transfer learning
- Conservative augmentation preserves features
- Large test set for robust evaluation

Model Architectures

Custom CNN

Baseline Architecture

Design:

- 3 Conv blocks
- 32 $\rightarrow$ 64 $\rightarrow$ 128 filters
- BatchNorm + ReLU
- MaxPooling
- FC + Dropout (0.5)

Training:

- 10 epochs
- Adam (lr=0.001)
- BCEWithLogitsLoss
- 2-3 min/epoch

Parameters:

- **2.1M** trainable

ResNet50

Transfer Learning

Design:

- Pretrained ImageNet
- Frozen backbone
- Custom 2-class head
- Deep residual blocks

Training:

- 5 epochs
- Adam (lr=1×10^{-4})
- Weight decay=1×10^{-4}
- CrossEntropyLoss
- 3-5 min/epoch

Parameters:

- **2K** trainable

EfficientNet-B2

Compound Scaling

Design:

- Compound scaling
- Depth + Width + Resolution
- Pretrained frozen backbone
- Efficient architecture

Training:

- 5 epochs
- Adam (lr=1×10^{-4})
- Weight decay=1×10^{-4}
- CrossEntropyLoss
- 4-6 min/epoch

Parameters:

- **7.8M** trainable

Model Comparison

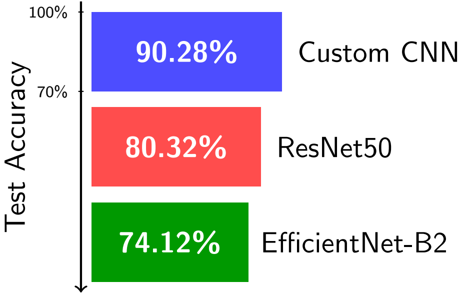
Model	Accuracy	Parameters	Epochs
Custom CNN	**90.28%**	2.1M	10
ResNet50	80.32%	2K	5
EfficientNet-B2	74.12%	7.8M	5

Custom CNN Classification Report

Class	Precision	Recall	F1-Score	Support
Real	0.91	0.90	0.91	8,061
AI-Generated	0.89	0.90	0.90	7,210
Accuracy			**0.90**	**15,271**

- **Winner:** Custom CNN at 90.28%
- **Surprise:** Baseline outperforms transfer learning
- **Balance:** Good precision/recall for both classes

Test Accuracy Comparison



Key Findings

- Custom CNN achieved highest accuracy
- Task-specific architecture beats generic transfer learning
- Longer training (10 vs 5 epochs) may have helped
- Transfer learning models may need fine-tuning

Conclusions & Recommendations

Model Selection Guide

Best Accuracy (90.28%)

Custom CNN

- Highest test accuracy
- Trained from scratch
- Task-specific features
- Recommended for this dataset

Speed & Efficiency

Custom CNN (2.1M params)

- Fastest inference
- Smallest model size
- Good for edge devices

ResNet50 (2K trainable)

- Minimal training overhead
- Moderate accuracy (80.32%)

Some Future Improvements

1. **Fine-tune transfer learning models**
   - Unfreeze later backbone layers
   - Train for more epochs
2. **Advanced augmentation**
   - CutMix, MixUp
   - Improve robustness
3. **Attention mechanisms**
   - Focus on synthetic artifacts
   - Add interpretability
4. **Expand training data**
   - More diverse AI generators
   - Cross-domain generalization

Deployment Recommendation

Production: Use **Custom CNN** for best accuracy (90.28%) on this specific task

Thank You!

Questions?

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Course: AAI-521 Computer Vision
Institution: University of San Diego
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Full report and code available at:
`github.com/marstonsward/AAI-521-Computer-Vision-Image-Classification-Project`